

QUICKLOAN MOBILE

Ethical Data Governance Review

Independent Data Governance Consultant Report

Prepared for	QuickLoan Mobile (Ghana)
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Classification	Confidential — Internal Use Only
Frameworks	Ghana DPA (Act 843) · Data Lifecycle · LARDI Metric

This report reviews QuickLoan Mobile's data pipeline against Ghana's Data Protection Act (Act 843), data governance principles, and fairness standards. It has three parts: a Governance Review Card, a Corrected Data Flow Diagram, and a Summary of the Review Process.

1 DELIVERABLE 1 — GOVERNANCE REVIEW CARD

Each finding below is assessed against Ghana's Data Protection Act (Act 843) and core data governance principles. The legal and bias risks need immediate action. The data quality risk can be fixed gradually without halting operations.

Findings at a Glance

#	Risk Area	Headline
1	Data Quality	No schema contract between app and database. High null rates, inconsistent phone formats, no validation gate. The model scores on guesses; rejections are unauditable.
2	Legal & Compliance SENSITIVE PII	Data collected before consent is captured. Contact-list harvesting takes third-party data with no lawful basis. Likely in breach of Act 843 before any loan is processed.
3	Bias & Fairness	GPS and device-type features act as demographic proxies in Ghana's economic geography. The model repeats past lending bias. No fairness audit before deployment.
4	Reporting Metric	LARDI — Loan Approval Rate Disparity Index. Worst group's approval rate divided by best group's rate. 0.80 threshold (the 4/5ths rule). Grouped bar chart, monthly.

Finding 1 · Data Quality Risk

Issue	There is no schema contract between the mobile app and the Raw Data DB. Income and employment fields are often missing. Phone numbers appear in three formats in the same column (+233XXXXXXXX, 0XXXXXXXX, and raw 10-digit). The API Gateway has no validation, so the Preprocessing Service takes whatever the app sends. <i>A model that fills in missing income is scoring on guesses. If a borrower disputes a rejection, QuickLoan cannot show what data drove the decision.</i>
Impact	Model outputs are only as good as the inputs. Missing values force the model to guess, which adds noise. Inconsistent formats cause silent errors. Both lower accuracy and weaken the audit trail.

Suggested Fix	Define a data contract for every model input — name, type, format, required or optional. Enforce it at the API Gateway: reject missing required fields and normalise phone numbers. Send bad records to a 72-hour review queue. Before each training run, check completeness. Fail the run if any required field drops below 95%.
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Finding 2 · Legal & Compliance Risk

Data Classification: **SENSITIVE**

Issue	The pipeline collects data — including the user's full contact list — before asking for consent. Under Act 843, processing personal data without a lawful basis is an offence, not a procedural gap. Consent must be freely given, specific, informed, and clear. Contact-list collection makes this worse: it captures personal data of third parties who never agreed and have no relationship with QuickLoan. There is no lawful basis for this. <i>Why Sensitive: the data includes financial history, precise GPS, device IDs, and a social graph. Each one qualifies as Sensitive PII under Act 843.</i>
Impact	QuickLoan faces liability in two directions: to users whose data is processed unlawfully, and to their contacts whose data is taken without their knowledge. The Data Protection Commission can issue enforcement notices, order deletion, and impose sanctions. QuickLoan also does not appear to be registered as a data controller, which Act 843 requires. It may be in breach before processing a single loan.
Suggested Fix	Three actions, in strict order of priority: 1. Register as a data controller with the Data Protection Commission now. Act 843 requires it; there is no grace period. 2. Stop collecting contact lists. No credit model needs them, and no consent can cover third-party contacts. 3. Add a consent module with purpose-specific opt-ins before any data leaves the app. Log every consent: timestamp, user ID, scope, version. Collect only the fields the model needs.

Finding 3 · Bias & Fairness Risk

Source of Bias	Historical lending data plus proxy variables (GPS, device type) that stand in for demographics.
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	The model uses GPS and device metadata as features. In Ghana, where someone lives is a strong proxy for income, education, and access to formal employment. Device type correlates with income tier and, in Ghana's market, with gender. A model trained on past approvals will repeat the biases of past loan officers. Without a fairness audit, there is no way to catch it. <i>Mobile lending promises financial inclusion in underserved markets. A model that penalises applicants for where they live or what phone they own does the opposite.</i>
Impact	Applicants from lower-income regions and women are more likely to get lower scores — not because they are riskier, but because the features track demographics. Act 843 prohibits processing that produces discriminatory outcomes on protected grounds. A rejected applicant who can show a pattern has a credible complaint with the DPC.
Suggested Fix	Before the next model deployment: Run a disparate impact analysis on current model outputs, split by gender and region. Remove GPS and device-type features until the audit is done. Add a pre-deployment fairness gate using equalised odds across groups. No model goes to production without passing. Send any rejection within 5 points of the threshold to human review. After deployment: Report approval rates by demographic segment every month. If any two groups differ by more than 20 points, convene a model review within 14 days.

Finding 4 · Storytelling / Reporting Recommendation

Metric	Loan Approval Rate Disparity Index (LARDI) The worst-performing demographic group's monthly auto-approval rate divided by the best-performing group's rate. Broken down by gender (male / female / other), age band (18–25, 26–35, 36–50, 50+), and region (Greater Accra vs. other regions).
Trigger Threshold	0.80 (the 4/5ths rule) If the LARDI falls below 0.80 for any segment, a model review is mandatory within 14 days.
Visualisation	Grouped Bar Chart Demographic group on the x-axis, approval rate (%) on the y-axis, bars grouped by month. A horizontal line marks the 80% threshold. Bars below the threshold are amber; bars needing immediate review are red.

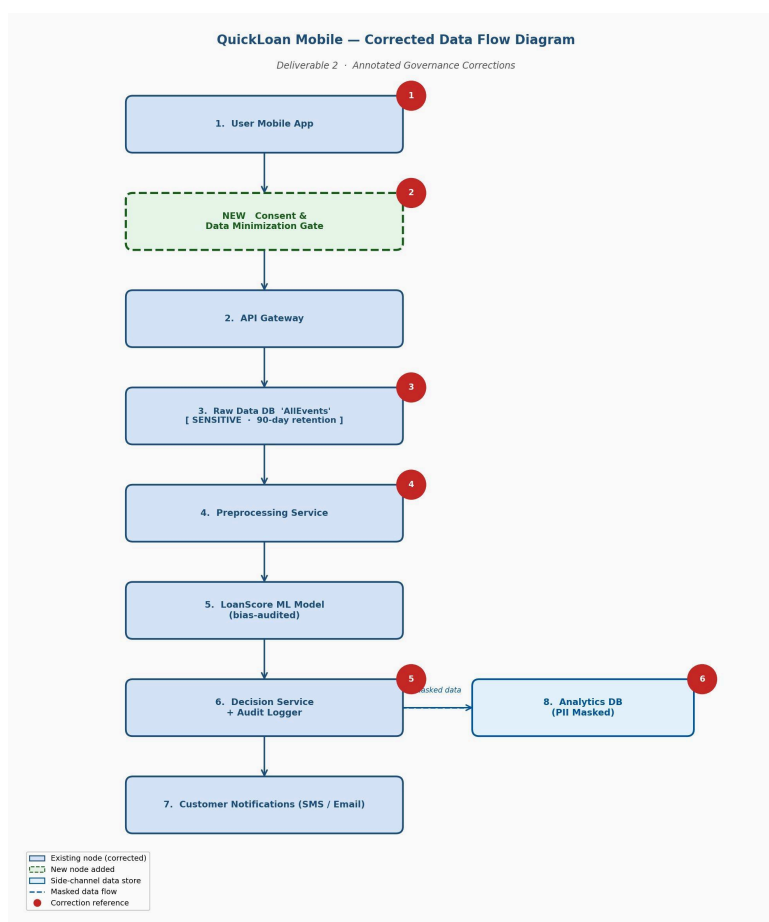
Why It Matters	Total approval rates look clean on a dashboard but hide who is actually being served. A single rate of 65% could mask 85% approval in one region and 40% in another. Without breaking the numbers down, neither the governance team nor a regulator can tell a fair model from one that systematically excludes a group.
Operating Cadence	Publish the LARDI monthly to the Data Governance Lead, Head of Risk, and the Board's Audit Committee. Report it quarterly to the Data Protection Commission as part of QuickLoan's data controller duties. Proactive disclosure builds accountability and lowers the risk of a surprise inquiry. Set automatic alerts: any month where LARDI falls below 0.80 for any segment opens a 14-day model review ticket.

All findings reference Ghana's Data Protection Act 2012 (Act 843) and the principles of Data Minimization, Purpose Limitation, Consent Management, Data Classification, and Retention. The 0.80 LARDI threshold follows the 4/5ths rule used in international fairness practice.

2 DELIVERABLE 2 — CORRECTED DATA FLOW DIAGRAM

The diagram below shows the corrected QuickLoan data pipeline. Each red badge marks a correction described in the annotation table that follows. The original pipeline had six core flaws — too much data collection, no consent, no classification, no PII handling, no decision transparency, and unmasked analytics — and each one is addressed with a specific change.

Corrected Pipeline



Correction Annotations

#	Pipeline Location	Original Flaw → Correction Applied	Why It Is Necessary
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1	Step 1 User Mobile App	Original flaw: App collects the full contact list, continuous GPS, and device logs — none of it relevant to credit scoring. Correction applied: Apply data minimization at collection. Stop collecting contact lists and device logs. Restrict location to region level. Collect only: name, national ID, stated income, loan amount, and repayment history.	Act 843 requires data to be adequate, relevant, and limited to what the purpose needs. Collecting a user's social graph for credit scoring has no legal or analytical basis.
2	NEW Step Consent & Data Minimization Gate	Original flaw: No consent step existed. Data went straight from the app to the API Gateway. Correction applied: Add a consent and minimization gate between the app and the API Gateway. It presents purpose-specific opt-ins, records the user's choices with timestamp and version, and blocks the flow if required consents are missing.	Act 843 requires consent to be freely given, specific, informed, and clear before processing starts. Consent must come before collection, not after.
3	Step 3 Raw Data DB 'AllEvents'	Original flaw: No classification labels. No retention policy. Records kept indefinitely with no purge schedule. Correction applied: Classify all records as SENSITIVE. Set a 90-day retention limit with automated purge and deletion logs. Tag each record with classification tier, consent ID, and deletion date. Restrict access to the Preprocessing Service only.	Classification determines security controls and retention. Act 843 requires data to be kept no longer than needed for its stated purpose.
4	Step 4 Preprocessing Service	Original flaw: No PII handling rules. Raw identifiers (name, phone, national ID) passed through to the feature set unchanged. Correction applied: Pseudonymize all direct identifiers before feature extraction. Replace name, phone, and national ID with stable internal tokens. Log every transformation. The ML model never sees raw PII.	The ML model has no need for a borrower's name or phone number. Passing raw PII into a scoring model widens the attack surface and makes re-identification easy if outputs leak.
5	Step 6 Decision Service	Original flaw: Auto-decisions issued with no audit log, no explanation, and no transparency to the borrower. A black box. Correction applied: Log every auto-decision: timestamp, model version, the top three contributing features in plain	Automated decisions without transparency or recourse are an ethical failure and a growing legal risk. Audit logs are the evidence base for any regulatory defence.

		language, outcome, and a plain-language explanation sent to the customer. Logs are immutable and kept for 3 years for regulatory audit.	
6	Step 8 Analytics DB	Original flaw: Raw PII stored without masking. Any analyst with read access could identify individual borrowers. Correction applied: Apply k-anonymity masking before records enter the Analytics DB. Remove direct identifiers; generalise quasi-identifiers. The Analytics DB holds no data that can re-identify a borrower.	Analytics databases usually have broader internal access than operational ones. Masking at the point of storage builds in privacy rather than relying on access controls alone.

Corrections reference Ghana's Data Protection Act 2012 (Act 843) and the principles of Privacy by Design, Data Minimization, Purpose Limitation, and Transparent Automated Decision-Making.

3 DELIVERABLE 3 — SUMMARY OF REVIEW PROCESS

The review began with a data lifecycle audit. I walked the pipeline stage by stage — collection, storage, processing, decision-making, disposal — and at each step asked whether the data was governed correctly for its sensitivity level.

Classification was the first tool. Once the dataset — financial history, GPS coordinates, a social graph from contact lists, device metadata — was identified as Sensitive PII under Act 843, the obligations followed: lawful basis, explicit consent, data minimization, retention limits, and security controls matched to the sensitivity tier. None of these were in place at intake, which meant QuickLoan had not classified its data at all. You cannot govern what you have not defined.

The lifecycle view then surfaced downstream effects. Unclassified data entering a preprocessing service with no handling rules stays unclassified at every later stage — including the analytics database, where it should have been masked. A decision service with no audit logging breaks accountability at the exact stage that matters most: the point where an automated system decides who gets credit.

The LARDI metric addresses a governance blind spot: total approval rates hide demographic gaps. A single rate of 65% could mask 85% in one region and 40% in another. By breaking rates down by gender, age, and region — and measuring them against the 4/5ths threshold — the LARDI turns fairness from a stated value into a measurable, auditable duty. Quarterly publication to the Data Protection Commission turns transparency from policy language into something a regulator can verify.

References: Ghana DPA (Act 843) · Data Lifecycle Framework · LARDI Metric.